

Certificate & Results Management System

Official Records Office · Tamper-proof certificates · semester-wise results · public verification · role-based admin portal

ABOUT · FEATURES · POLICIES

1 · ABOUT THE APPLICATION

The university runs a secure certificate platform combining a **public website** (verify a certificate by hash, check a student result, raise an issue — no login, no PII exposed) and an **admin portal** (dashboard, certificates, results, analytics, issues, activity log, user management, branding and downloads). It is built on a Node.js + Express REST API with a PostgreSQL database and a React front end. Certificate hashes are optionally anchored on the Sepolia testnet blockchain for permanent, public proof.

2 · KEY FEATURES

 **Tamper-Proof Certificates**
SHA-256 hash per certificate; optional IPFS + Sepolia on-chain anchoring.

 **Instant Public Verification**
Verify by hash only; validity status returned, never personal data.

 **Role & Department Scoping**
7 roles with per-user section permissions and department-level API enforcement (403).

 **Results Management**
Semester-wise results with multiple subjects; unique roll + semester prevents duplicates.

 **Bulk Upload**
CSV / XLSX with header detection and mismatch report before committing.

 **Complete Audit Trail**
Every key action logged with search; sessions survive restarts.

3 · HOW IT WORKS

1
 **Certificate is Issued**
The records office issues a certificate; its unique hash is written to the blockchain.

2
 **Stored Securely**
The fingerprint stays permanently linked to the student record on-chain.

3
 **Verified Instantly**
Employers and students verify authenticity in seconds — no paper, no third party.

4 · ROLES & ACCESS

ROLE	DEFAULT ACCESS
Admin	Full access to every section & all departments.
Normal User	Section-based access, department-scoped.
Department Head	Certificates, results, verify, analytics, issues, activity.
Privacy Manager	Verify, issues, activity, downloads.
Cyber Security	Monitor, verify, issues, activity.
HR / Legal	Verify, issues, activity, downloads.

5 · TECHNOLOGY STACK

Frontend	React 18 (Create React App), light/dark themes
Backend	Node.js + Express REST API
Database	PostgreSQL 16 (JSONB for subjects, permissions & departments)
Blockchain	Optional Sepolia testnet anchoring
Files	XLSX parsing, CSV export
Ops	PM2 process manager, Docker Compose

6 · POLICIES

 **PRIVACY POLICY**

- ▶ **Public verification only.** A hash returns only "valid" / "not found" — names, roll numbers, results and contact details are never exposed publicly.
- ▶ **No unnecessary collection.** Only data required to operate the portal is stored (name, roll number, course, department, year, email, marks/grades, optional mobile/Aadhaar for exports).
- ▶ **Under your control.** Results are visible only to administration and staff granted access; department restrictions further limit visibility.
- ▶ **Share the certificate, not the record.** Holders share the hash so employers verify authenticity without revealing private records.

 **SECURITY & DATA PROTECTION**

- ▶ **Strong password storage:** scrypt hashing; legacy SHA-256 hashes upgraded automatically at login.
- ▶ **Session-based auth:** random 64-char tokens in PostgreSQL; 12-hour idle expiry.
- ▶ **Rate limiting & lockout** on repeated failed logins per IP.
- ▶ **Server-side authorization** on every protected endpoint (session + role + permission + department).
- ▶ **ROPA & DPIA:** apps complete a GDPR Art. 30 register and Art. 35 risk assessment before going live / before high-risk processing; the admin portal tracks compliance per application.

7 · SUPPORT & HOW TO RAISE AN ISSUE

Students can raise an issue directly through the portal (roll number + semester); the records office triages it as open / resolved / rejected. Administrative and technical support is handled by the University Official Records Office.